Information Collection Document - Cinema Booking System

In this document, we gathered information on cinema booking web applications and collected specific information about functionalities, use cases and ideas for our own implementation.

We studied 3 booking web applications, **Village Cinemas, Odeon SA and AMC Theaters**.

We excluded any information and display of advertisements and concessions (food, drinks etc) since they are irrelevant and out of scope for this assignment.

Overview

1. Village Cinemas

Topbar contains Movies, Cinemas, Venues in order to filter by each category. This way a user can either search by movie, availability and proximity (for example a user that lives near a cinema can look up if a movie is available there). Its interface is designed for quick navigation, with prominent Tickets buttons overlaying each movie poster. To complete a booking, a user typically selects a film, chooses a specific cinema location and screening time, and then proceeds to a seating map to pick their spots.

A button named Tickets, redirects us to the booking system. There we choose a movie, venue, and afterwards we have a calendar with available showtimes.

2. Odeon Cinemas

The UI only contains movies. We cannot search by venue. When we press a movie we can see the available showtimes, and a popup to buy a ticket. The booking process is designed to be as few clicks as possible: users select a film from the View Cinema Listings menu, choose their preferred seat on an interactive map, and pay for the tickets.

3. AMC Cinemas

More similar to Village Cinemas, we can search by movie theatre. The main page includes large posters with the movies and after we click on a movie we can see the button Tickets. The booking journey is highly integrated: once a showtime is selected, the system prompts the user to log in to AMC Stubs for points, followed by a reserved seating selection.

The common theme in all the booking web applications is the main page includes large posters with the movies, a topbar or sidebar with options for venues, cinemas or movies. When we choose a movie, we can select a showtime with a small calendar. So the flow most of these applications use is Movie - Select Showtime - Buy tickets.

On a more technical note, we create a table to compare the booking systems.

Functionality	Village Cinemas	Odeon	AMC Cinemas
Movie Posters	Yes	Yes	Yes
Interactive Seat Selection	Yes	Yes	Yes
Authentication System	Yes	Yes	Yes
Mobile Responsiveness	Yes	Yes	Yes
Email for ticket verification	Yes	Yes	Yes
QR for check in	Yes (on ticket)	Yes (on ticket)	Yes (on ticket)

This creates a very clear implementation and design plan for our own application. The differences between these applications mostly lie on UX choices and not implementation and architectural differences.

Functionalities Analysis

The basic functionalities that must be:

1. Authentication System
The system must be able to authenticate users and separate users based on roles. Guests (unauthenticated users) must be able to just see movie posters and showtimes. Authentication must be easy, and the user can select between SSO access (Google etc) and normal registration and access. Authenticated users must have the ability to make reservations. Admins must be able to change/add/remove venues, movies, showtimes, and ticket validators must have the ability to check in users with the generated QR codes on their tickets.

2. Reservations
The system must offer the ability to make reservations, either singular or more tickets. The payment system must be easy to use and in a safe transactional environment (will be simulated in the web application). The user must be able to easily select a movie, venue and seats based on an interactive seat plan.

3. Check in system
The system must offer the ability to easily authenticate users and allow the ticket validators to check if the tickets are valid.

4. UI/UX

The system must offer an easy to use, intuitive interface to allow the user to navigate the movies, see trailers, descriptions, age limitations, and choose venues, cinemas and the seat they want based on the seating plan. The seating plan must always show only the available seats (unavailable seats must be another color so it's clear it’s not available).

Use Case Descriptions

Use Case 1: Basic Reservation

Actor: Authenticated User

Steps:

1. The user logs in / registers into the application.
2. Chooses the movie he wants to see
3. Chooses venue/cinema and showtime
4. The system displays the available seats
5. The user chooses the seats he wants to reserve
6. The user pays for the seats and the system stores the information in the database
7. The system sends a verification email with PDF tickets that include the QR code for easy entry into the venue

Use Case 2: Entry Validation

Actor: Ticket Validator

Steps:

1. The guest goes to the venue to see the movie he reserved
2. The guest finds the reservation email on his phone and prepares the QR codes
3. The ticket validator, using his phone, scans the ticket/QR code on the guest’s phone
4. The ticket validator’s phone prints a success message. The system updates the database and the list of users that have checked in is updated
5. The guest enters the venue successfully and the system logs the process

Use Case 3: New Movie Addition (compatible with venue, cinema addition)

Actor: Administrator

Steps:

1. The administrator chooses the movies page
2. He adds a new movie, along with its description, trailer link etc

3. He chooses the venue, cinema it will be played and the dates of showtime
4. He clicks save, updating the database and the system with the new movie information
5. The users can now choose, see and make reservations with the new movie

Use Case 4: Showtime edit (compatible with venue, time edit)

Actor: Administrator

Steps:

1. A problem with a venue has come up and the administrator wants to change the datetime of a movie show.
2. He enters the system with his credentials in the showtimes page
3. He chooses the movie, venue and time of the movie
4. He changes the time of the show. The system checks that it doesn't conflict with another showtime
5. If everything is correct, the system is updated, updating the database with the new showtime
6. If there are reservations for the movie, an automatic email is sent to the users that have made reservations informing them of the new time of the movie

Use Case 5: Authentication

Actor: User

Steps:

1. The user wants to have an account in the system to easily make reservations
2. He clicks register and is redirected to the registration page
3. He enters his credentials and email and confirms. The system is updated, and adds the user to the database.
4. He receives an onboarding email with a link that directs him to the sign in page

Use Case 6: Checked in users lookup

Actor: Ticket Validator / Administrator

Steps:

1. The actor wants to see how many users have entered the venue / show
2. He enters in the system with his credentials
3. He goes to the showtime page and clicks on users
4. He is able to see all the users that are already checked in and the users that have made a reservation but have not yet checked in